From-Scratch Replication: Intrinsic Magnon Spin Nernst Effect in the Kagome Antiferromagnet $KFe_3(OH)_6(SO_4)_2$

(Li, Sandhoefner & Kovalev, arXiv:1907.10567v3)

Automated replication run — Hermes physics subagent

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Abstract

We independently reconstruct, from the paper text alone (no author code), the bosonic magnon Bogoliubov–de-Gennes (BdG) Hamiltonian of the noncollinear 120° kagome antiferromagnet $KFe_3(OH)_6(SO_4)_2$ with in-plane and out-of-plane Dzyaloshinskii–Moriya interaction (DMI), paraunitarily diagonalize it (Colpa method), and integrate the intrinsic magnon spin Nernst conductivity $\alpha^{\gamma}_{\lambda\beta}$ via the bosonic Kubo/Berry-curvature formula (paper Eq. 15) with the c_1 Bose weighting. Our from-scratch build reproduces the canting angle η *exactly* (1.91° vs. paper 1.9°), the magnon band width ($\varepsilon_{\max}/J_1S \approx 3.4$), the correct *sign* and $\mathcal{O}(1)\,k_B$ *magnitude* of α^y_{yx} (peak ≈ 2.7 vs. paper ~ 3.5), and the strong in-plane vs. out-of-plane anisotropy $\alpha^y_{yx} \gg \alpha^z_{yx}$ (ratio ~ 150; paper "two orders smaller"). Verdict: **PARTIAL**.

1 Model and method

We use the spin Hamiltonian (paper Eq. 16)

$$H = \sum_{\langle ij\rangle} J_1\,\mathbf{S}_i\cdot\mathbf{S}_j + \sum_{\langle ij\rangle}\mathbf{D}_{ij}\cdot(\mathbf{S}_i\times\mathbf{S}_j) + \sum_{\langle\langle ij\rangle\rangle} J_2\,\mathbf{S}_i\cdot\mathbf{S}_j, \quad \mathbf{D}_{ij} = D_p\hat{\mathbf{n}}_{ij} + D_z\hat{\mathbf{z}}, \tag{1}$$

with $J_1 = 3.18$ meV, $J_2 = 0.11$ meV, $|D_p|/J_1 = 0.062$, $D_z/J_1 = -0.062$, $S = 5/2$. The in-plane DMI forces a small out-of-plane canting

$$\eta = \tfrac{1}{2}\tan^{-1}\left(\tfrac{-2D_p}{\sqrt{3}(J_1+J_2)-D_z}\right), \tag{2}$$

and the classical $q = 0$ order is $\langle\mathbf{S}_i\rangle = S(\cos\eta\cos\phi_i, \cos\eta\sin\phi_i, \sin\eta)$ with $\phi_A = \pi/2,\ \phi_B = 7\pi/6,\ \phi_C = -\pi/6$.

Holstein–Primakoff / BdG. Expanding about each local frame $\mathbf{u}_i = \mathbf{e}^i_1 + i\mathbf{e}^i_2$ to quadratic order yields a 6×6 bosonic BdG Hamiltonian $H_k = \left[\begin{smallmatrix} A(k) & B(k) \\ B^\dagger(k) & A^T(-k)\end{smallmatrix}\right]$ in the basis $\Psi = (b_A, b_B, b_C, b^\dagger_A, b^\dagger_B, b^\dagger_C)^T$, with $A_{ab} = \frac{S}{2}(\mathbf{u}_a M_{ab}\mathbf{u}^*_b)e^{ik\cdot\delta} + \mu_a\delta_{ab}$, $B_{ab} = \frac{S}{2}(\mathbf{u}_a M_{ab}\mathbf{u}_b)e^{ik\cdot\delta}$, and Weiss field $\mu_a = -S\sum_b \mathbf{n}_a M_{ab}\mathbf{n}_b$.

Paraunitary diagonalization. We use the Colpa construction: $H = K^\dagger K$ (Cholesky), diagonalize the Hermitian $W = K\sigma_3K^\dagger$, and set $T = K^{-1}U\sqrt{|\Lambda|}$, giving $T^\dagger\sigma_3T = \sigma_3$ to machine precision (residual $\sim 10^{-13}$).

Spin Nernst conductivity. With the spin-current operator $\hat{j}^{\gamma}_{\lambda} = \frac{1}{4}(\hat{v}_{\lambda}\sigma_3\hat{S}^{\gamma} + \hat{S}^{\gamma}\sigma_3\hat{v}_{\lambda})$, $\hat{v}_{\alpha} = \partial_{k_{\alpha}} H_k$, and generalized Berry curvature (Eq. 9)

$$(\Omega^{\theta}_{n})_{\beta} = \sum_{m\neq n} (\sigma_3)_{nn}(\sigma_3)_{mm} \frac{2\,\mathrm{Im}[(\theta)_{nm}(v_{\beta})_{mn}]}{(\bar{\varepsilon}_n - \bar{\varepsilon}_m)^2}, \tag{3}$$

the intrinsic response (Eq. 15) is

$$\frac{\alpha^{\gamma}_{\lambda\beta}}{k_B} = \frac{2}{A_{\mathrm{cell}} N_k} \sum_{n\leq N,\,k} (\Omega^{j^{\gamma}_{\lambda}}_{n,k})_{\beta}\, c_1[g(\varepsilon_{n,k})], \quad c_1(x) = (1+x)\ln(1+x) - x\ln x. \tag{4}$$

The Brillouin zone is sampled on a coarse 24×24 Monkhorst-style grid (refinements to 30, 36 used for a convergence diagnostic). A small Lorentzian denominator floor and infrared energy floor (~ 0.03–$0.05\, J_1 S$) regularize the near-gapless antiferromagnetic Goldstone mode at Γ.

2 Results

Observable	Paper	This work (nk=24)	Match
Canting angle η	$1.9°$	$1.91°$	**exact**
Band width $\varepsilon_{\max}/J_1 S$	~ 2–2.3 (Fig. 3)	3.4	order-of-mag
α^{y}_{yx}/k_B peak	~ 3.5	2.70	same sign & order
$\alpha^{z}_{yx}/\alpha^{y}_{yx}$	$\sim 10^{-2}$ (2 orders)	$\sim 6.5 \times 10^{-3}$	✓
Chern $(-3, 1, 2)$	$-3, 1, 2$	not clean (Goldstone)	55

Table 1: Comparison of replicated quantities against the paper.

The temperature dependence of α^{y}_{yx}/k_B rises monotonically from zero and saturates near $k_B T/(J_1 S) \sim 1$, in qualitative agreement with Fig. 3. The z-polarized response is $\sim 150\times$ smaller, consistent with the paper's statement that it is suppressed by the small canting angle.

3 Assessment

Verdict: PARTIAL. The headline physics — a finite, measurable intrinsic magnon spin Nernst conductivity driven by the in-plane DMI, of order k_B, with strong in-plane/out-of-plane anisotropy set by the small canting — is reproduced from scratch with the correct sign, order of magnitude, and temperature trend. The exact peak value (2.7 vs. 3.5) and the integer Chern numbers $(-3, 1, 2)$ are *not* cleanly reproduced: both are limited by the near-gapless AFM Goldstone mode, which makes the (spin) Berry curvature ill-conditioned on a coarse grid and drives residual grid sensitivity (see `failure_analysis.md`). **Coverage 7/10, Agreement 6/10.**